



GARASU

BETA

GLASS AGING, RELAXATION AND SIMULATION UTILITY

User Guide



Beta

Learn

Lab

Fitter

Data Analysis

GARASU

Glass Aging, Relaxation And Simulation Utility Beta

Project **none** ▾ New project

Every glass carries the **thermal fingerprint** of how it was made, written in its fictive temperature. GARASU simulates and fits that history, using the heterogeneous **Takeda-Lucas (TL)** model alongside the **Tool-Narayanawamy-Moynihan (TNM)** model and **RelaxPy**, an implementation of the **Mauro-Allan-Potuzak (MAP)** model. It handles differential scanning calorimetry (DSC), modulated DSC (MDSC), enthalpy, volume and annealing data.

Start learning Analyze data Open Fitter Download for Mac / Windows

Tools

Learn

Experiments & models

How structural relaxation is measured, and what the TNM, MAP and Takeda-Lucas models assume, with live simulations throughout.

- Annealing, Kovacs asymmetry and memory, DSC, enthalpy recovery, MDSC (C_p^* , reversing/non-reversing and their caveats)
- The mathematics of TNM, MAP (RelaxPy) and TL
- Mean field vs Prony vs domain T_f : see the distributions

[Open Learn](#)

Data Analysis

Your data

Turn raw DSC heat flow and relaxation data into normalized curves, fictive temperatures and enthalpy.

- Glass and liquid baselines $\rightarrow C_p^N, T_f(T), T_f'$ for heating and cooling scans
- Heating after annealing: aligned scans, heat-flow difference ΔH_f , recovered enthalpy $\Delta H(t_g)$, relaxation function ϕ and a Kohlrausch-Williams-Watts (KWW) fit
- Volume / density / length during annealing: stretched-exponential fit

[Open Data Analysis](#)

Fitter

Step by step

Program any thermal history, load your measurements, then compute with chosen parameters or fit them.

- Ramps, anneals with log-spaced times, T-jumps, MDSC modulation
- DSC, MDSC C_p^*/C_p^* , enthalpy, volume, T_f , annealing data
- Bounded local / global fits, errors, correlations, Akaike and Bayesian information criteria (AIC, BIC)

[Open Fitter](#)

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In a hurry?

The separate **Quick Start** document takes you from installing to a first result in a few minutes. This guide explains every screen and option in detail.

1 • What GARASU does

Every glass carries a thermal fingerprint of how it was made, written in its fictive temperature T_f . GARASU simulates that history, analyzes your measurements, and fits relaxation models to them.

GARASU contains four tools. The red bar on the left of every page switches between them, on the web and in the desktop app; the icon at its top returns to the home page.

Tool	Use it to...
Learn	understand the experiments (annealing, Kovacs jumps, differential scanning calorimetry or DSC, enthalpy recovery, modulated DSC or MDSC) and the three models described below, with live simulations on every topic.
Lab	try a cool-heat scan, an anneal or a T-jump interactively and compare the three models side by side.
Fitter	program any thermal history, load your data, then compute with chosen parameters or fit the parameters, with uncertainties.
Data Analysis	turn raw DSC heat flow into normalized C_p , fictive temperatures and recovered enthalpy, and fit volume (or density, length...) relaxation during annealing.

The three models

Model	In one line
Takeda–Lucas (TL)	Heterogeneous domains, each with its own Vogel temperature and local fictive temperature $T_{f,i}$; the spread of $T_{f,i}$ persists in equilibrium, giving temperature-dependent nonexponentiality and enthalpy fluctuation (J. Chem. Phys. 160, 174504, 2024).
Tool–Narayanaswamy–Moynihan (TNM)	One fictive temperature for the whole sample, nonlinearity x and a stretched-exponential Kohlrausch–Williams–Watts (KWW) kernel.
Mauro–Allan–Potuzak (MAP) / RelaxPy	Nonequilibrium viscosity with Prony-series components (Wilkinson, Mauro & Mauro, SoftwareX 2018).

Your data stay on your computer

All calculations run locally, in your browser or in the desktop app. Nothing you load is uploaded anywhere.

Beta software

GARASU is in beta: features, defaults and occasionally results may change between versions. Record the version number (bottom of the home page in the desktop app) with your results.

2 • Getting GARASU

2.1 In the browser

Open wtakeda-research.github.io/GARASU. Nothing to install; it works in Chrome, Edge, Firefox and Safari, and always shows the latest version. After the page has loaded, the tools also work offline.

2.2 Desktop app

Download the installer from github.com/wtakeda-research/GARASU/releases/latest, or click **Download for Mac / Windows** on the home page.

macOS (Apple silicon and Intel)

- 1 Download `GARASU-<version>-universal.dmg`.
- 2 Open it and drag **GARASU** into **Applications** (choose **Replace** if asked).
- 3 Open GARASU. The first time, macOS says it cannot verify the developer: click **Done**.
- 4 Open **System Settings** → **Privacy & Security**, scroll down and click **Open Anyway**, then confirm.

This is needed once per computer, because the app is signed with the group's own certificate rather than an Apple Developer ID.

Windows 10 / 11

- 1 Download `GARASU-Setup-<version>.exe`.
- 2 Run it. If Windows shows “Windows protected your PC”, click **More info** → **Run anyway**.
- 3 Choose the install folder (no administrator rights are needed) and finish.
- 4 Start GARASU from the Start menu or the desktop shortcut.

2.3 Updates

The desktop app checks for a new version when it starts and every six hours; you can also check with **GARASU** → **Check for Updates...** (Mac) or **Help** → **Check for Updates...** (Windows). A new version downloads in the background and the app offers **Restart now** to install it.

If macOS cannot install an update in place, the app says so and offers **Download**, which opens the release page: install the new disk image (.dmg) as in 2.2. The web version needs no updates.

2.4 One window for every tool

The desktop app opens a single window on the home page. The bar on the left switches tools inside that window, and each tool stays as you left it: a computed Fitter run is still there after a visit to Learn. Shortcuts: `⌘/Ctrl 1` Learn, `2` Lab, `3` Fitter, `4` Data Analysis, `⌘/Ctrl 0` home. The version number is shown at the bottom of the home page.

3 · Controls shared by every tool



Figure 3.1. Header of the Fitter. Top row: tool name, project, contents and settings; second row: the four steps. Data Analysis has the same layout.

Control	What it does
Tool bar (left)	Learn, Lab, Fitter and Data Analysis; the icon at the top opens the home page. The current tool is highlighted.
Project ▼ and Contents	The project you are working in, shared by the Fitter, Data Analysis and the home page (chapter 8).
Temperature K / °C	Unit for every temperature on screen: plots, thermal-history inputs, T_g and its bounds, data limits. Values are stored in kelvin, so switching never changes a calculation.
Time axis auto / linear / log	<i>Auto</i> (default): real (linear) time whenever the temperature changes with time (ramps, MDSC), so scans are shown in real time; log time only for isothermal histories (jump + hold). Relaxation during holds and time-domain hold data stay on log time. <i>Linear</i> or <i>log</i> force one scale for every time axis.
☾	Switches between the light (default) and dark colour scheme.
Export ▼ (on each plot)	Saves that plot as a PNG or JPEG image, an SVG vector graphic, or its curves as CSV (section 8.6).
Hover on a plot	Shows the nearest data point and its values.

Typing numbers
Inputs accept any number format (`1e5` , `0.5` , `-40`). Temperatures are read in the unit selected in the header.

4 • Learn

Learn is a self-contained course on structural relaxation. The contents on the left jump to any section; **Part I** covers experiments and **Part II** the models.

Learn

TL TNM MAP Experiments Models

Contents

- 1 · Fictive temperature
- Part I · Experiments
 - 2 · Isothermal annealing
 - 3 · Kovacs: asymmetry, memory
 - 4 · DSC cooling and heating
 - 5 · Enthalpy recovery
 - 6 · MDSC and C_p
- Part II · Models
 - 7 · TNM
 - 8 · MAP / RelaxPy
 - 9 · Takeda–Lucas
 - 10 · See the difference
 - 11 · Side by side
 - 12 · Further reading

Learn: glass relaxation, from experiment to model

Part I covers the experiments used to study structural relaxation: annealing, Kovacs jumps, differential scanning calorimetry (DSC), enthalpy recovery and modulated DSC (MDSC). Each one comes with a live simulation. Part II explains the mathematics of the three models and where they differ. The Tool–Narayanawamy–Moynihan (TNM) model gives the whole sample one fictive temperature. The Mauro–Allan–Potuzak (MAP) model, as implemented in RelaxPy, splits the relaxation into Prony components, and all of them return to the same temperature in equilibrium. The Takeda–Lucas (TL) model gives every domain its own fictive temperature, so the fictive temperatures stay spread out even in the equilibrium liquid.

1 Fictive temperature and structural relaxation

Above the glass transition, a liquid's structure keeps pace with temperature. On cooling, the structural relaxation time τ grows so quickly that the structure eventually cannot keep up and freezes. Tool described the frozen structure by a **fictive temperature** T_f ; the temperature at which the liquid would have this structure in equilibrium. In equilibrium $T_f = T$. In the glass, T_f records the thermal history, which is why GARASU calls it the *thermal fingerprint*.

Enthalpy and volume follow T_f . With liquid and glass heat capacities $C_{p,l}$ and $C_{p,g}$, the normalized heat capacity measured by DSC is

$$C_p^N(T) = \frac{C_p - C_{p,g}}{C_{p,l} - C_{p,g}} = \frac{dT_f}{dT}.$$

Relaxation towards equilibrium has three well-known signatures, and every model below has to reproduce all three:

- **Nonexponential.** After a small temperature step, the response follows a stretched exponential (Kohlrausch–Williams–Watts, KWW), $\Psi(t) = \exp[-(t/\tau)^\beta]$ with $0 < \beta \leq 1$.
- **Nonlinear.** τ depends on the structure (T_f) as well as on T . A glass therefore relaxes faster when it approaches equilibrium from above than from below (Kovacs asymmetry).
- **Non-Arrhenius.** In equilibrium, $\tau(T)$ follows a Vogel–Fulcher–Tammann (VFT), Adam–Gibbs or Mauro–Yue–

Figure 4.1. Learn. The switch at the top (TL / TNM / MAP) chooses the model used by all simulations in Part I.

Part I • Experiments

Each experiment has its protocol and equations, then a live simulation of the same example glass, selenium ($T_g = 308.13$ K, $m = 64.14$, with the TL parameters of Takeda & Lucas 2024; for MAP the glassy viscosity is Arrhenius with $B = x m T_g$, $x = 0.5$, and $C = 0$):

- **Isothermal annealing:** relaxation function $\phi(t)$ after a 10 K down-jump, compared with a linear KWW curve, and the apparent KWW parameters.
- **Kovacs experiments:** asymmetry of approach (± 5 K jumps) and the memory effect (Kovacs; Macedo–Napolitano crossover).
- **DSC on cooling and heating:** heating scans after five cooling rates and Moynihan's cooling-rate plot (apparent $\Delta h/R$ and fragility).
- **Enthalpy recovery:** heating scans after annealing and $\Delta H(t_a)$.
- **MDSC:** total, reversing and non-reversing signals, C'' , and how the reversing step moves with the modulation period (the Hutchinson and Schawe caveats).

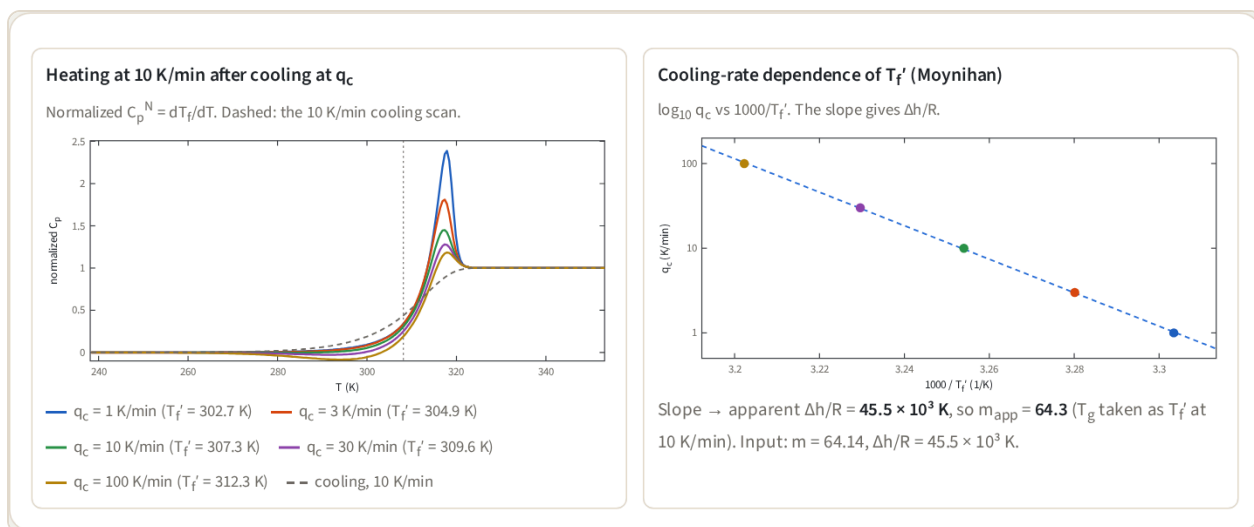


Figure 4.2. DSC section: heating after five cooling rates (left) and the cooling-rate dependence of T_f' (right).

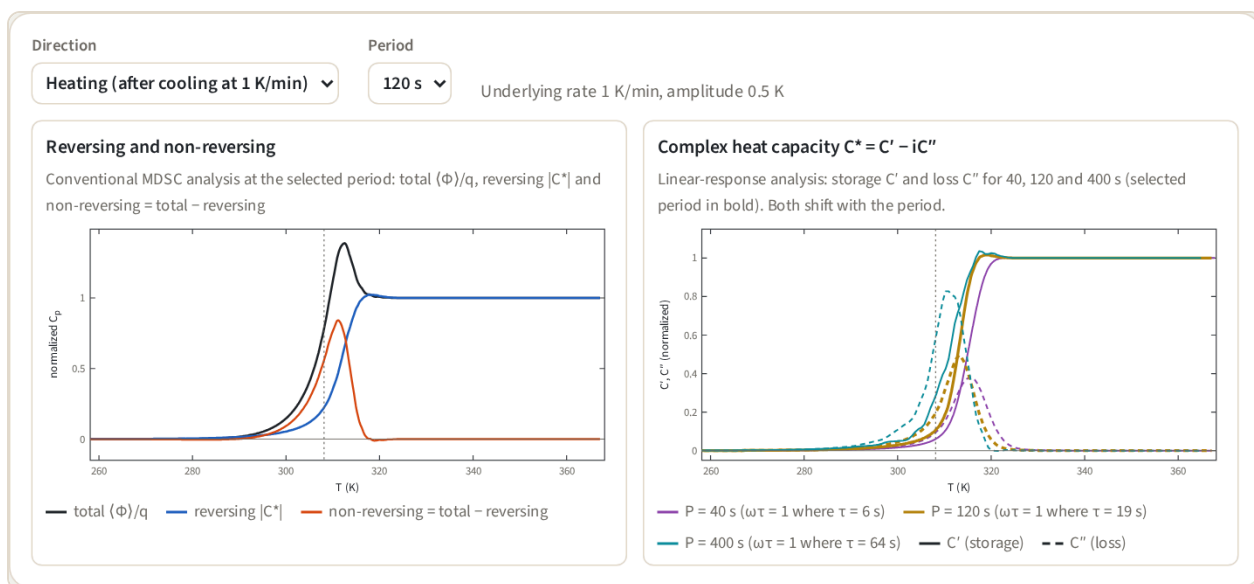


Figure 4.3. MDSC section. Choose **Direction** (heating or cooling) and **Period**. Left: the conventional analysis (total, reversing $|C^*|$ and non-reversing). Right: the complex heat capacity, C' (solid) and C'' (dashed), for 40, 120 and 400 s.

Part II • Models

The mathematics of TNM, MAP and TL, what each assumes, and an interactive comparison of where the fictive temperatures sit.



Figure 4.4. “See the difference”: the three models run the same cool–anneal history. Drag the slider or press ► Play; change the cooling rate, annealing temperature and TL width f .

5 • Lab

The Lab (formerly Explorer) is a quick, always-live playground: change a number and the plots update. Use it to build intuition; use the Fitter for your own histories and data.

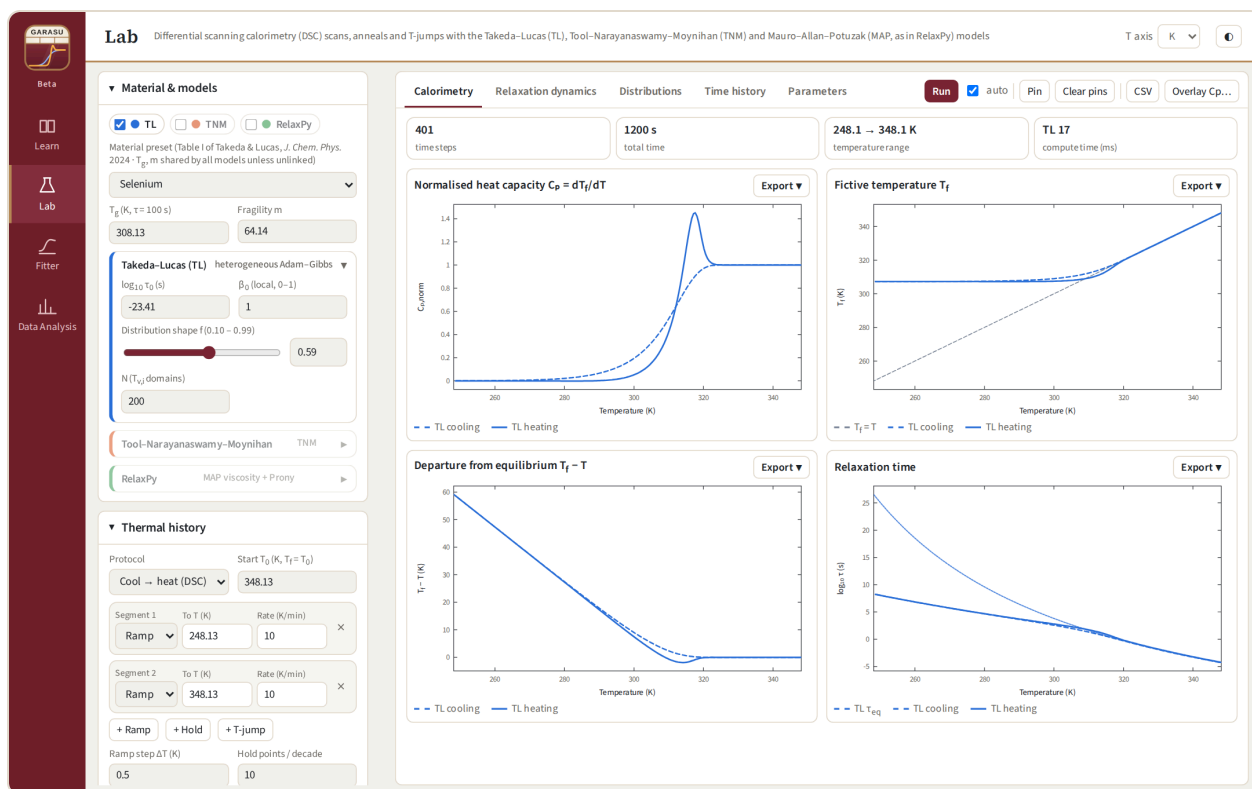


Figure 5.1. Lab: models and material on the left, protocol below; tabs on the right.

- **Models:** tick TL, TNM and/or RelaxPy. **Material preset** loads the TL parameters of Table I of Takeda & Lucas, *J. Chem. Phys.* 2024; T_g and m are shared by the models unless unlinked in their sections.
- **Thermal history:** choose a protocol (cool $\rightarrow$ heat, slow cool, anneal, Kovacs T-jump) or build segments with + Ramp, + Hold, + T-jump.
- **Tabs:** Calorimetry (C_p , T_f), Relaxation dynamics, Distributions (TL relaxation-time distributions and $\beta_{KWW}(T)$), Time history, Parameters.
- **Run / auto:** untick **auto** to update only when you click **Run**. **Pin** keeps a run for comparison; **CSV** exports the curves.

6 • Fitter

The Fitter works in four steps, shown as a switch in the header. You can move between them at any time.

Step	What you do	Right panel shows
1 Thermal history	Build the temperature program.	T(t) and the time steps
2 Experimental data	Optional: add measurements.	your data and the linked segment
3 Model & parameters	Choose a model, values, which parameters are free, and bounds.	history and data
4 Compute & fit	Click Compute or Run fit .	all results

Nothing is calculated until you click

Steps 1–3 only show the thermal history and your data. The model runs when you click **Compute** or **Run fit** in step 4. Any later change clears the old result and asks you to compute again, so a result on screen always matches the current settings.

6.1 Step 1 • Thermal history

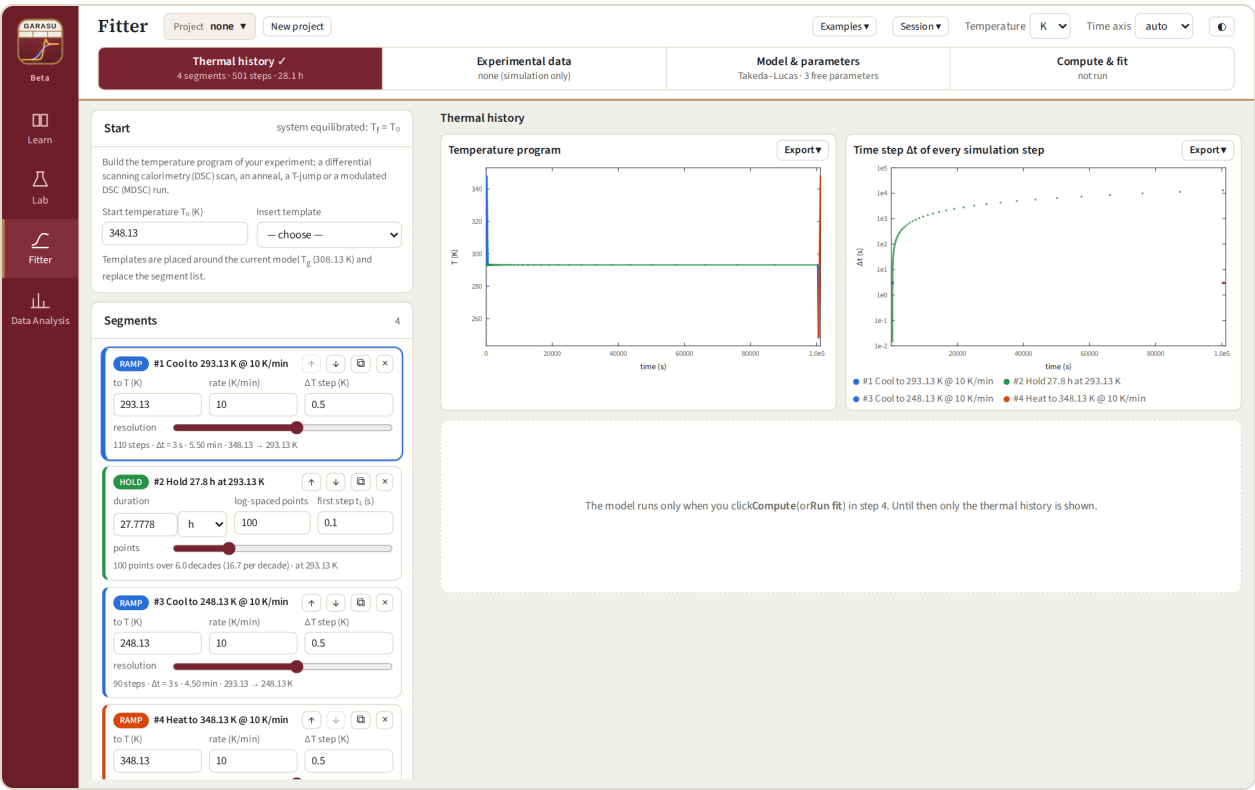


Figure 6.1. Step 1 with the template “Aging below T_g , then DSC heating”.

- 1 **Start temperature T_0 .** The sample starts in equilibrium at T_0 ($T_f = T_0$). Choose T_0 above T_g .
- 2 **Insert template** (optional). Templates are placed around the current model T_g , so set T_g in step 3 first if your material differs from the default.
- 3 **Add segments with + Ramp, + Hold / anneal, + T-jump, + MDSC.** Reorder, duplicate or delete them with the small buttons on each card.

Segment	Inputs	Notes
Ramp	target T, rate (K/min), ΔT step	0.1–0.5 K steps are accurate for DSC; the slider sets ΔT on a log scale.
Hold / anneal	duration (s, min, h, d), log-spaced points, first step t_1	Points are spaced logarithmically from t_1 to the duration.
T-jump	target T	Instantaneous change (e.g. Kovacs experiments).
MDSC	target T, underlying rate, amplitude $\pm A$, period, points per period	Rate 0 gives quasi-isothermal MDSC with a duration.

Under the segments, the summary shows the number of simulation steps and the total time. The right panel plots the temperature program and the time step of every simulation step.

6.2 Step 2 • Experimental data (optional)

The screenshot shows the GARASU Fitter interface. The top bar includes tabs for 'Thermal history', 'Experimental data', 'Model & parameters', and 'Compute & fit'. The 'Experimental data' tab is active, showing two datasets: 'enthalpy relaxation during aging (synthetic)' and 'DSC heating C_p raw (synthetic)'. The 'Add dataset' form is open, showing options for data type, linked history segment, and paste/file options. The 'First rows' table displays data points for x (s) and y.

x (s)	y
3	108.402
4.62993	108.401
7.14542	108.397
11.0276	108.349
17.019	108.353
26.2656	108.356
40.536	108.289
62.5596	108.256

Figure 6.2. Step 2 with two datasets and the Add dataset form open.

- 1 Click **+ Add dataset** and choose the **Data type** (table in section 11).
- 2 Choose the **Linked history segment**: the part of the thermal history the data belong to (e.g. the heating ramp for a DSC heating scan, the hold for annealing data).
- 3 **Paste / file**: paste two or more columns or click **Load file...**, then pick the x and y columns and units. Or **Synthesize from model** to generate test data with noise.
- 4 Optional: use $x \geq / \leq$ to limit the fitted range, a **weight**, a **y multiplier**, and **fit an offset and a scale factor** ($y = a + b \cdot \text{model}$) when the data are not normalized.
- 5 Click **Add dataset**. Tick or untick a dataset to include it;  edits it, $\times$ removes it.

Raw data are fine

Types such as “ C_p or heat flow vs T”, “Enthalpy vs T” and “property vs t” fit their baselines or linear coefficients exactly at every step, so you can load raw measurements. To normalize DSC data yourself first, use Data Analysis (section 7).

6.3 Step 3 · Model and parameters

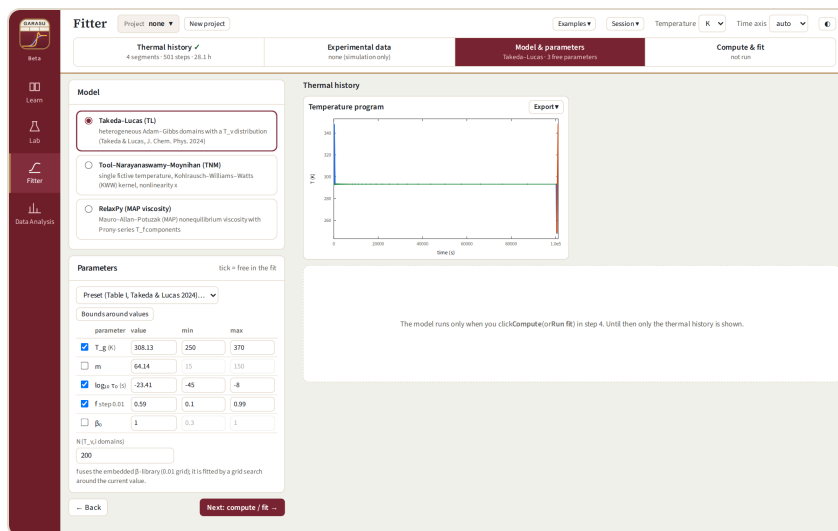


Figure 6.3. Step 3: model, parameter values, tick = free in the fit, and fit bounds.

- Choose **Takeda–Lucas**, **TNM** or **RelaxPy (MAP)**.
- Type values. Ticked parameters are **free** in a fit and move between **min** and **max**; unticked ones stay fixed. **Bounds around values** sets sensible bounds.
- TL: **Preset** loads Table I of Takeda & Lucas, *J. Chem. Phys.* 2024; **N** is the number of domains (200 is accurate). **f** is fitted on its 0.01 grid.
- TNM: choose the exact KWW summation or the Prony kernel for display (fits always use Prony).
- MAP: **RelaxPy example glass**; A can follow from continuity at T_g , and $p = 0.3082 \cdot m$ (RelaxPy default).

6.4 Step 4 · Compute and fit

Compute with chosen parameters

Forward calculation, no fitting. Check or adjust the values and click **Compute**. With data, the agreement (R^2 and root-mean-square error, RMSE, per dataset) is shown.

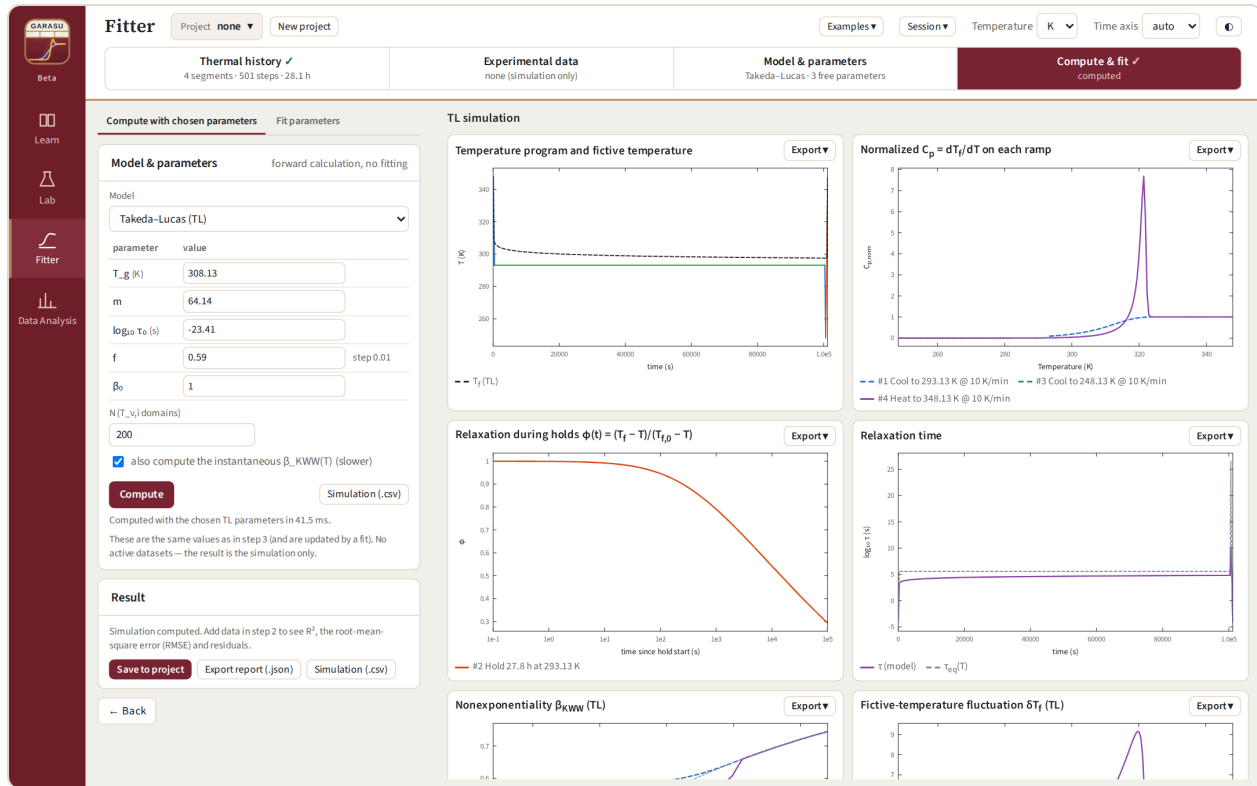


Figure 6.4. Result of **Compute**: temperature program and T_f , normalized C_p on each ramp, relaxation during holds, and relaxation time.

Extra results for the TL model

- Fictive-temperature fluctuation δT_f vs temperature, with the equilibrium value.
- Map of the local fictive temperatures $T_{f,i}$ vs time, rebinned on a uniform T_f grid (probability density in K^{-1}), with $T(t)$ and $\langle T_f \rangle(t)$.
- $\beta_{KWW}(T)$: tick also compute the instantaneous $\beta_{KWW}(T)$ (adds about a second).

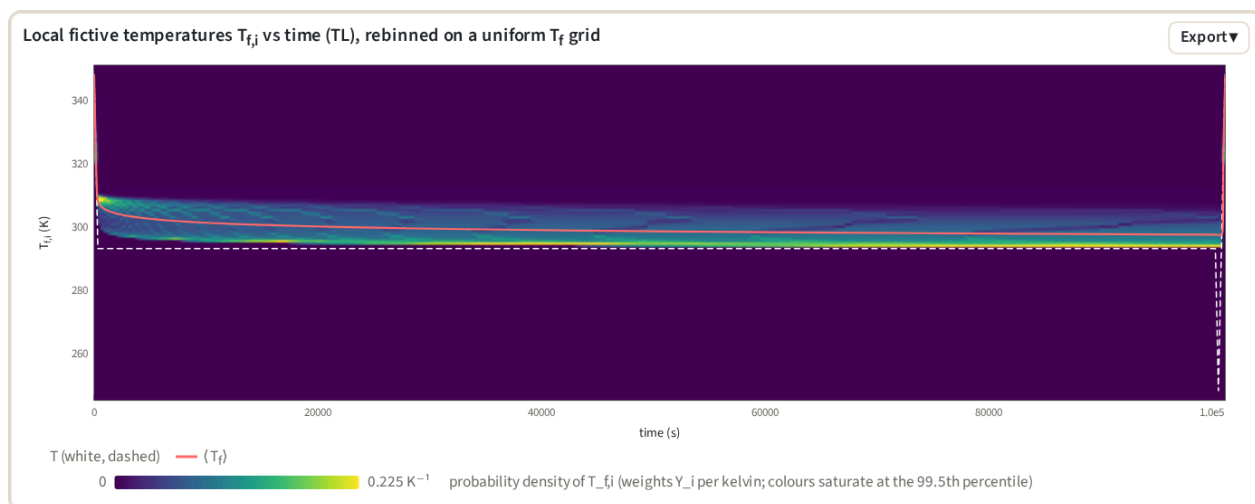


Figure 6.5. TL: local fictive temperatures during cooling, aging and reheating.

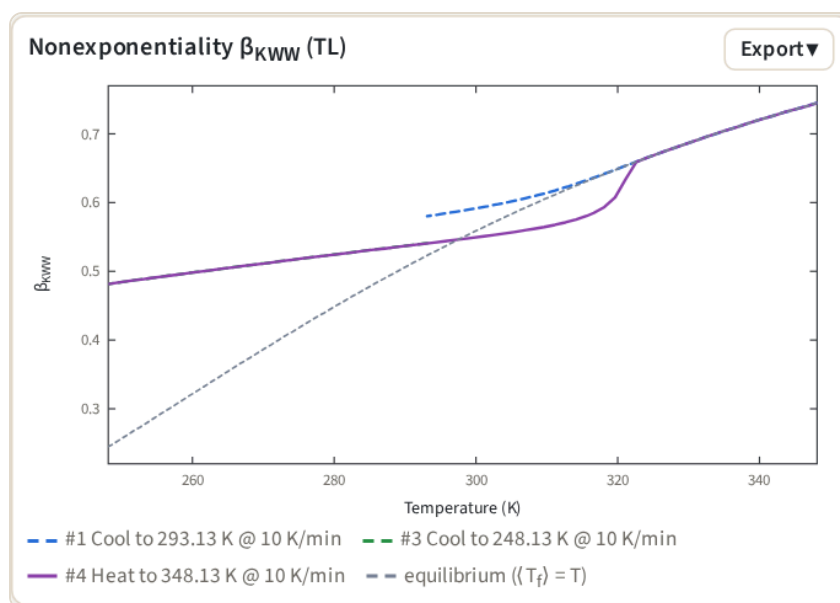


Figure 6.6. TL: instantaneous β_{KWW} along each segment and in equilibrium (grey).

Fit parameters

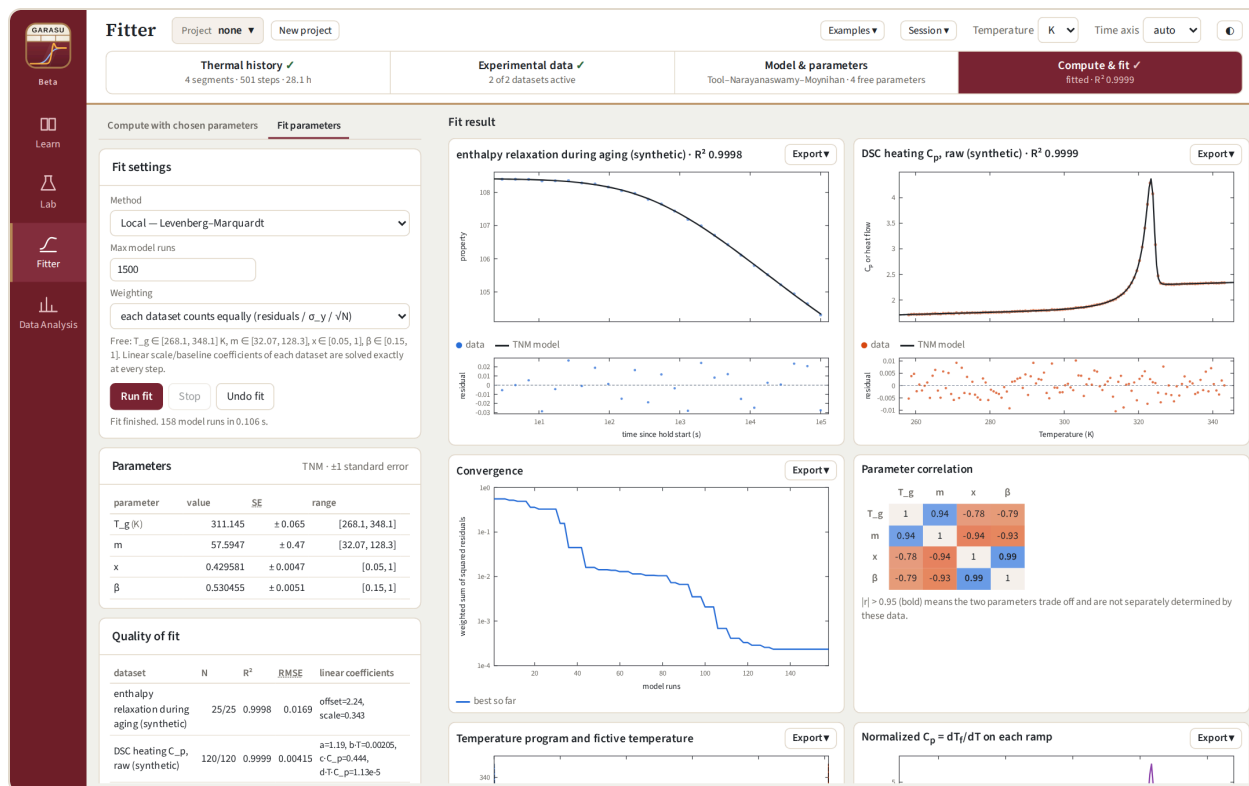


Figure 6.7. Fit result: data with the fitted model and residuals, convergence, parameter correlation, and the parameter table with standard errors.

- 1 Open the **Fit parameters** tab (needs at least one active dataset and one free parameter).
- 2 Choose the **Method**: *Local - Levenberg-Marquardt* (fast, starts from the current values) or *Global - differential evolution + LM polish* (slower, searches the whole box).
- 3 Set **Max model runs** and the **Weighting** (each dataset equal, or each point equal).
- 4 Click **Run fit**. **Stop** ends early; **Undo fit** restores the previous values.

The result lists each parameter with ± 1 standard error, flags parameters *at bound* or *poorly determined*, and gives R^2 , reduced χ^2 , and the Akaike and Bayesian information criteria (AIC, BIC). In the correlation matrix, $|r| > 0.95$ (bold) means two parameters trade off and are not separately determined by your data.

Exports

Export report (.json) (parameters, errors, statistics, history), **Model vs data (.csv)**, **Simulation (.csv)** (t , T , T_f , τ , τ_{eq} , segment), and **Export ▼** on each plot. **Save to project** stores the whole result in the open project (section 8.3).

6.5 Sessions and examples

Session ▼ holds the Fitter's own state: *New session*, *Save session file (.json)* (history, data, model, values, bounds, fit options) and *Open session file...* Session files from earlier versions ("project" files) open the same way. To keep data and fits of one study together, use a project (chapter 8). **Examples ▼** loads ready-made cases: a selenium DSC cycle, aging + DSC with synthetic enthalpy data, and MDSC with synthetic C_p'/C_p'' .

7 · Data Analysis

Data Analysis prepares your own measurements. It has three tabs, shown as a switch in the header. Each tab has **Load example data** (synthetic selenium data made with the TL model, parameters of Se from Takeda & Lucas 2024) so you can try everything first.

7.1 Heat-flow scans

Add a heat-flow run

File (.csv, .txt, instrument export)

Choose File

No file chosen

...or paste columns

Time (min),Temperature (°C),Heat Flow (W/g)

0.00,-120.0,-0.1000

0.05,-119.5,-0.1000

0.10,-119.0,-0.1000

0.15,-118.5,-0.1000

...

161 numeric rows, 3 columns detected.

Temperature column

Heat-flow column

Temperature unit

Heat-flow unit

Sign convention

Scan

Rate |q| (K/min)

Annealing time t_a (s)

Annealing T_a (K)

Name

Add run

Cancel

Figure 7.1. Adding a run: paste or load a file, then check the detected columns and units.

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- 1 Click **+ Add run**, then choose a file (.csv, .txt, instrument export) or paste columns.
- 2 Check the **Temperature** and **Heat-flow** columns and their units (K or °C; mW, W/g or arbitrary). Column names like “Temperature (°C)” and “Heat Flow (W/g)” set these automatically.
- 3 Set the **Sign convention**: *endothermic up*, or *endothermic down* (*exo up*) as in TA Instruments exports.
- 4 **Scan** is detected from the temperature trend (heating/cooling). Enter the **rate** and, for mW, the **sample mass**.
- 5 For a heating scan measured after annealing, enter the **annealing time** t_a and T_a . Leave them empty for the unaged reference.
- 6 Click **Add run**.

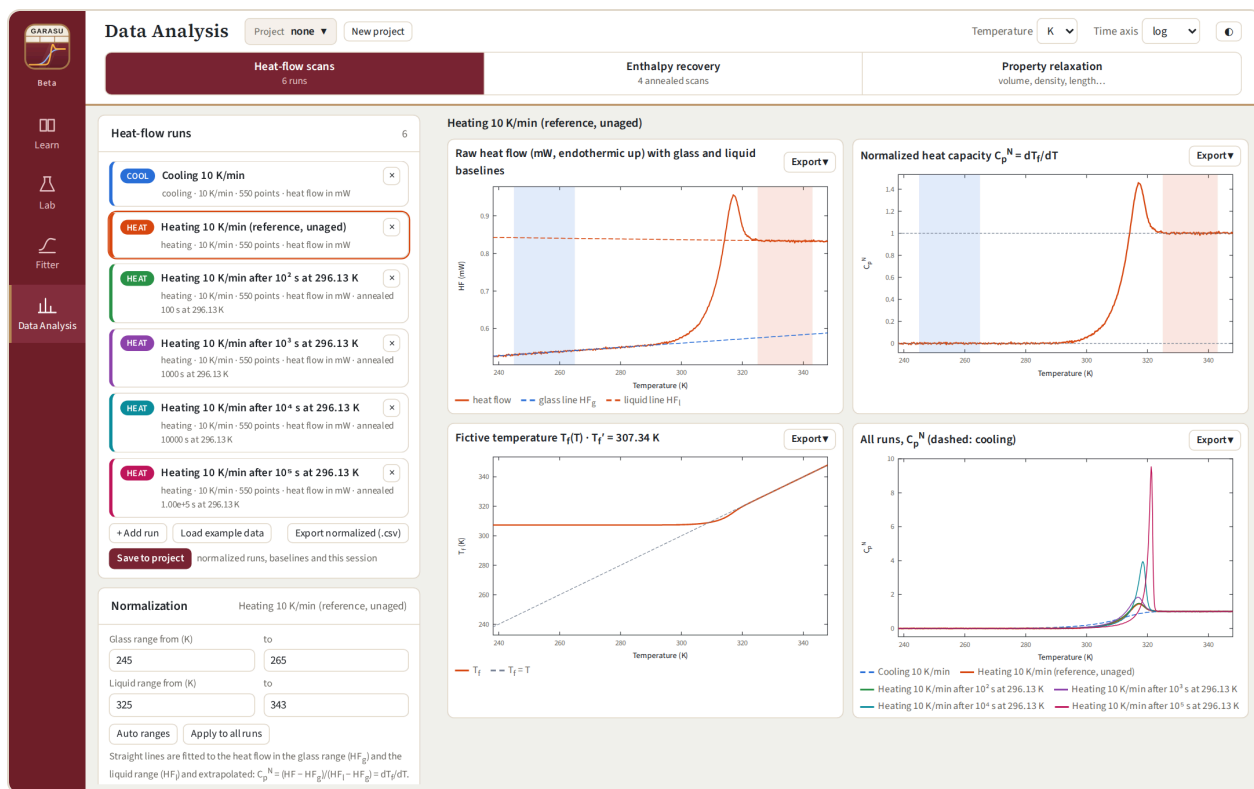


Figure 7.2. Normalization of the reference heating scan. Blue band: glass range; orange band: liquid range; dashed: the fitted baselines.

Normalization. Enter a **glass range** and a **liquid range** (or **Auto ranges**). A straight line is fitted to the heat flow (HF) in each range and extrapolated, giving

$$C_p^N = (HF - HF_g) / (HF_l - HF_g) = dT_f/dT, \quad T_f(T) = T^* - \int_T^{T^*} C_p^N dT'$$

with T^* at the start of the liquid range (Moynihan's area matching). The card reports the limiting fictive temperature T_f' , the midpoint ($C_p^N = 0.5$) and ΔC_p at T_f' . **Apply to all runs** copies the ranges; **Export normalized (.csv)** saves T , HF , C_p^N , T_f for every run; **Save to project** stores them, with both baselines, in the open project (section 8.2).

Choosing the ranges

- Glass range: well below the transition, where the scan is straight; avoid the start-up transient.
- Liquid range: above the overshoot, where the scan is straight again.
- For comparisons (cooling rates, aging series), use the same ranges for all runs.

7.2 Enthalpy recovery

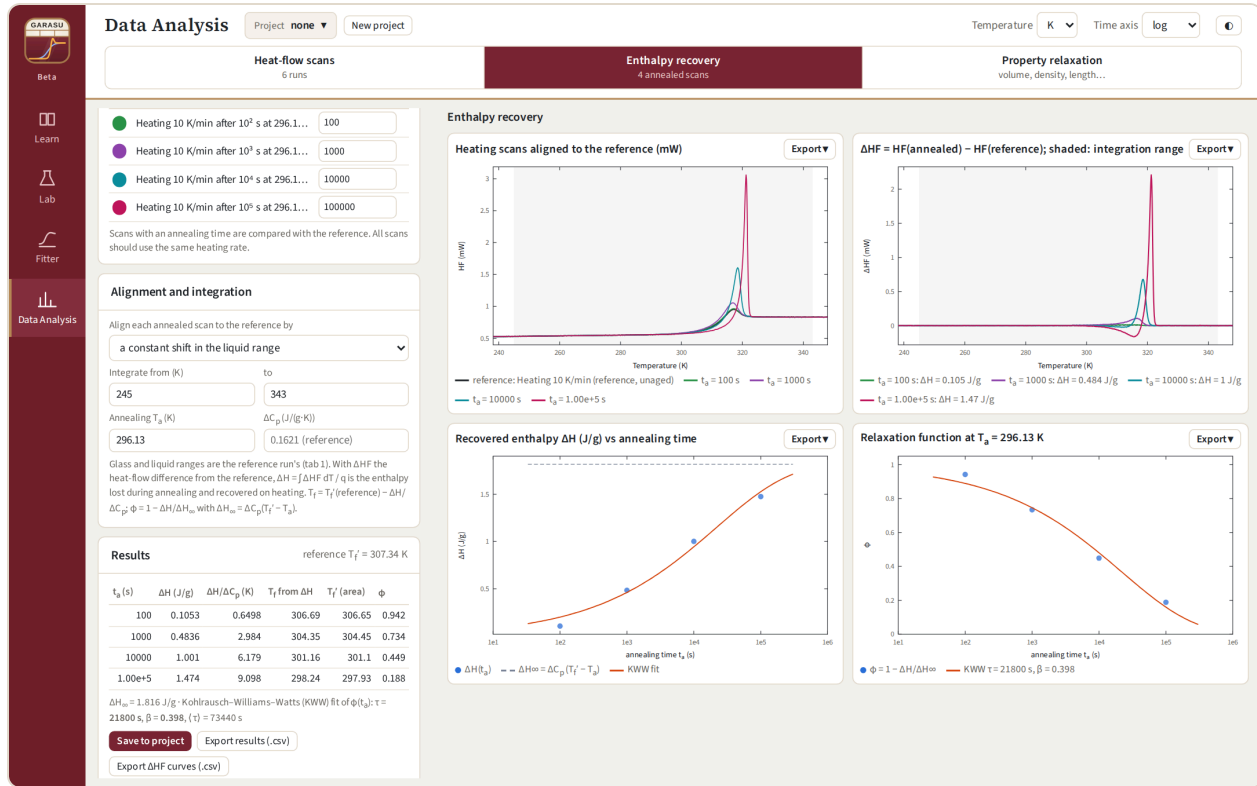


Figure 7.3. Enthalpy recovery: aligned scans, ΔHF(T), recovered ΔH vs annealing time, and φ(t_a) with a KWW fit.

- 1 Choose the **Unaged reference** (a heating scan without annealing).
- 2 Enter or check the annealing time t_a of each annealed heating scan.
- 3 Choose how each annealed scan is **aligned** to the reference: a constant shift in the liquid range, or a straight line through the glass and liquid ranges (use this if the glass lines differ slightly).
- 4 Check the **integration range** (default: from the glass range to the liquid range) and enter T_a.
- 5 Optionally override ΔC_p (default: the reference at T_f).

$$\Delta H = (1/q) \int \Delta HF dT, \quad T_f = T_f'(\text{ref}) - \Delta H/\Delta C_p, \quad \phi = 1 - \Delta H/\Delta H_{\infty}, \quad \Delta H_{\infty} = \Delta C_p(T_f' - T_a)$$

Result column	Meaning
ΔH	enthalpy lost during annealing and recovered on heating (J/g with mW + mass or W/g; mJ with mW and no mass)
$\Delta H/\Delta C_p$	the same in kelvin: the drop of the fictive temperature during annealing
T_f from ΔH / T_f' (area)	fictive temperature after annealing, from ΔH and independently from area matching of the annealed scan (they should agree)
ϕ	relaxation function; with three or more annealing times a KWW fit gives τ , β and $\langle \tau \rangle$

Use the same heating rate

ΔH is computed with the reference's rate. All scans should be heated at the same rate and ideally measured on the same sample.

7.3 Property relaxation

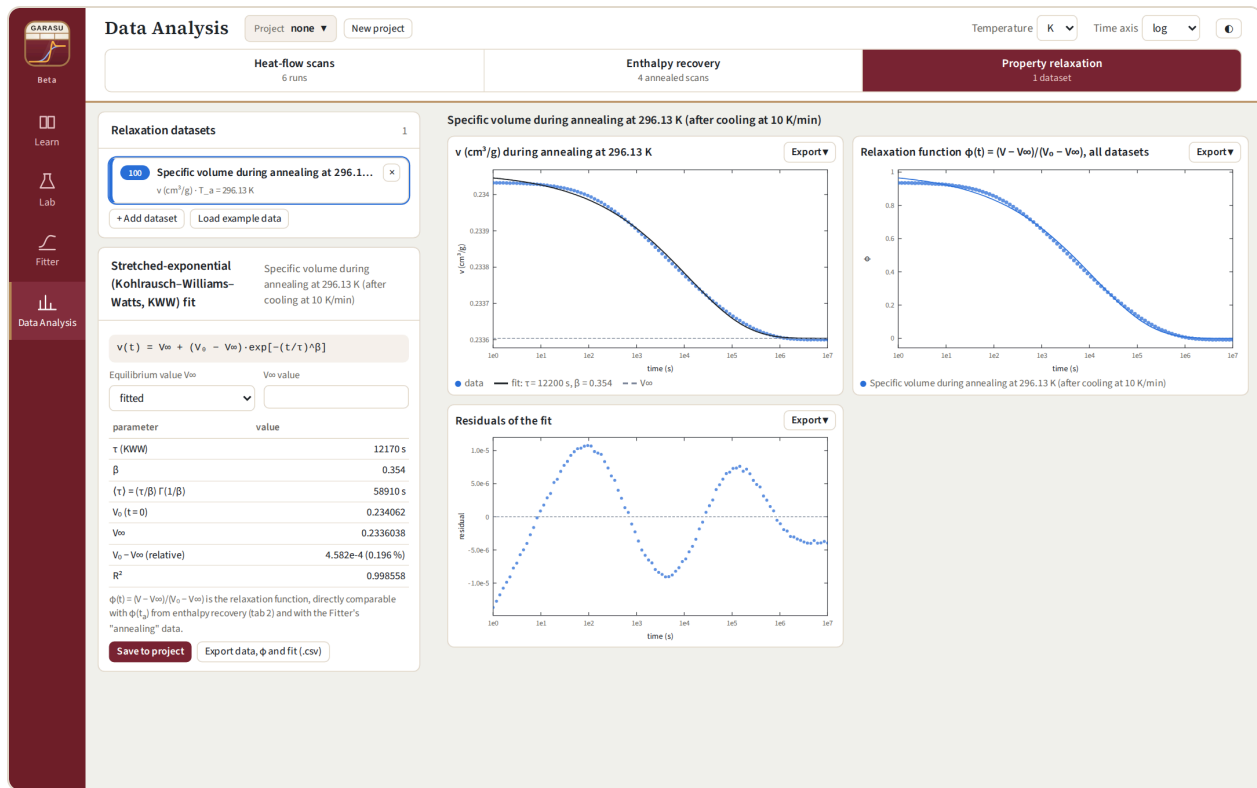


Figure 7.4. Specific volume during annealing with the stretched-exponential fit, $\phi(t)$ and residuals.

Add a dataset of a property against time during isothermal annealing: volume, density, length, refractive index... (+ Add dataset: pick the time column and unit, the property column, a label and T_a). The fit is

$$V(t) = V_{\infty} + (V_0 - V_{\infty}) \exp[-(t/\tau)^{\beta}]$$

with V_∞ **fitted** or **fixed** at a value you enter (use fixed when the data stop well before equilibrium). The table gives τ , β , $\langle \tau \rangle = (\tau/\beta)\Gamma(1/\beta)$, V_0 , V_∞ , the relative change and R^2 . Several datasets can be compared on the $\phi(t)$ plot.

Systematic residuals (as in Figure 7.4) are expected for real glasses: relaxation is nonlinear, so a single stretched exponential is only an approximation. Fitting the model in the Fitter accounts for this.

7.4 From analysis to fit

The simplest route is a project: save the normalized runs (tab 1) and choose **Fit in the Fitter** in Contents (section 8.3). Alternatively, export the normalized curves (tab 1) or $\phi(t)$ (tabs 2, 3) as CSV and paste them into Fitter step 2 as “DSC — normalized C_p ” or “Annealing — normalized relaxation $\phi(t)$ ”, linked to the matching segment of a thermal history that reproduces your experiment.

8 • Projects

A project keeps one study together: the normalized scans from Data Analysis, the fits made from them, and the sessions of both tools. A typical project follows a series of DSC measurements from raw heat flow to fitted model parameters.

Saved item	Contents	Saved from
Normalized heat capacity	T , raw HF, C_p^N , $T_f(T)$; T_f' , midpoint, ΔC_p ; scan direction and rate, annealing t_a and T_a ; the glass line $HF_g = a + b \cdot T$ and liquid line HF_l with their ranges	Data Analysis, tab 1
Enthalpy recovery	ΔH , T_f , $\phi(t_a)$ for each annealed scan, and the KWW fit	Data Analysis, tab 2
Property relaxation	each dataset with its stretched-exponential fit	Data Analysis, tab 3
Fit result	model, all parameters (value, free or fixed, ± 1 standard error, bounds), correlation matrix, R^2 , χ^2 , AIC, BIC; thermal history; every dataset with the model curve on the same points, its linear coefficients, and the baselines of its source scan	Fitter, step 4
Session	the complete state of the tool, to continue later	both tools

Where projects are kept

Projects are saved on this computer, in the browser (web) or in the app (desktop), and appear in every tool and on the home page. The web and desktop versions, and different browsers, keep separate projects: use **Export project file** to move one between them or to back it up (section 8.5).

8.1 Create a project

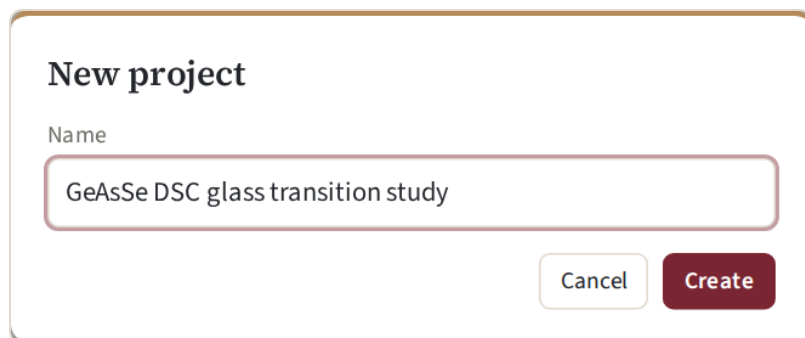


Figure 8.1. New project.

Click **New project** in the header of Data Analysis, the Fitter or the home page, and give it a name, for example *GeAsSe DSC glass transition study*. The project stays open in every tool until you close it or open another. The **Project ▼** menu opens, renames, exports, imports, closes and deletes projects.



Figure 8.2. The project menu.

8.2 Save normalized data

- 1 In Data Analysis, add your heat-flow runs and set the glass and liquid ranges (section 7.1).
- 2 Click **Save to project**. Each run is saved as a normalized heat capacity with its baselines, together with the session. Saving again replaces the earlier copy of the same run.
- 3 Tabs 2 and 3 have their own **Save to project** buttons for enthalpy recovery and property relaxation.

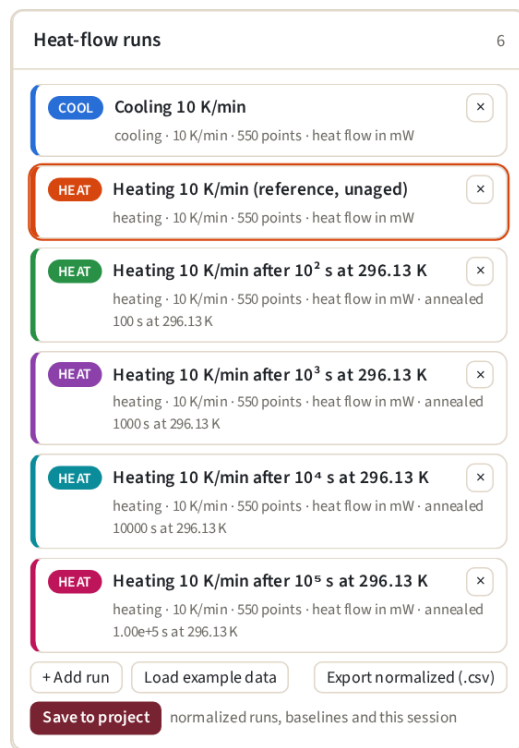


Figure 8.3. Data Analysis, tab 1: **Save to project** below the runs.

8.3 Fit from the project

- 1 Open **Contents** and click **Fit in the Fitter** on a normalized run (from Data Analysis) or **Add as dataset** (in the Fitter). Or, in step 2, click + **Add dataset** and choose the **From project** tab.
- 2 Keep **build the thermal history of this scan** checked: the Fitter writes the history from the saved scan (cooling, the annealing hold at T_a for t_a , cooling below the data, and the heating scan at its rate) and links the dataset to the scan. Further runs are linked to their own scans.
- 3 Choose the model and free parameters (step 3), then **Run fit** (step 4).
- 4 Click **Save to project** in the results and give the fit a name.

Add dataset [X]

Data type
DSC — normalized C_p (dT_f/dT) vs T [v]

Linked history segment
#2 Heat to 348.13 K @ 10 K/min [v]

Paste / file Synthesize from model **From project**

Normalized run from "GeAsSe DSC glass transition study"
Heating 10 K/min after 10^4 s at 296.13 K [v]

heating · 10 K/min · $T_f = 301.1$ K · annealed 10000 s
Glass line $HF_g = 0.3852 + 5.920e-4 \cdot T$ over 245–265 K; liquid line $HF_l = 0.88539 - 1.556e-4 \cdot T$ over 325–343 K. These travel with the dataset into the saved fit.
build the thermal history of this scan: cool to 296.13 K, hold
☒ 10000 s, cool to 238.2 K, heat to 348 K at 10 K/min (replaces the current history)

use $x \geq$ (K) use $x \leq$ weight
all all 1

name
Heating 10 K/min after 10^4 s at 296.13 K

☐ fit an offset and a scale factor for this dataset ($y = a + b \cdot \text{model}$)

x limits are in K for temperature data and seconds (from the start of the hold) for annealing data.

Add dataset

Figure 8.4. Fitter, step 2: a normalized run from the project, with its baselines.

The fit result keeps the fitted curve against the experiment point by point, all model parameters with their errors, the quality of fit, and the glass and liquid lines HF_g and HF_l of each scan used. Download it from Contents as CSV (data and model columns, with the parameters and baselines in the header) or JSON (everything).

Parameters searched on a grid (such as f in TL) are free but have no standard error; the CSV marks them as grid-searched.

8.4 Contents

Contents lists everything saved in the open project, newest first. Each item has an action (**Open here**, **Add as dataset**, **Fit in the Fitter**), **CSV** and **JSON** downloads, and **×** to delete it. **Open here** on a fit result restores its history, data, model and parameters in the Fitter.

8.5 Back up and share

Project ▼ → **Export project file** saves the whole project as one `.garasu.json` file. **Import project file...** adds it on another computer, in another browser, or in the desktop app. Export regularly: clearing the browser's site data removes projects saved in that browser.

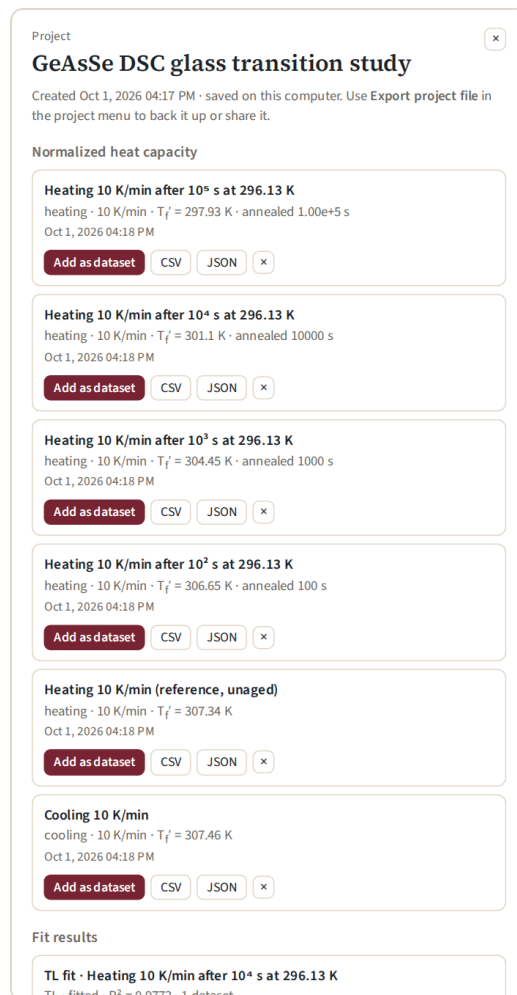


Figure 8.5. Contents of a project, grouped by type.

8.6 Export a plot

Every plot in Lab, the Fitter and Data Analysis has **Export ▼**:

Format	What you get
PNG / JPEG image	the plot at three times screen resolution, with its title and legend, on a white background
SVG (vector)	the same figure as a vector graphic, to edit in Illustrator or Inkscape
CSV data	every curve of the plot as x, y column pairs, with the title and axis labels in the header

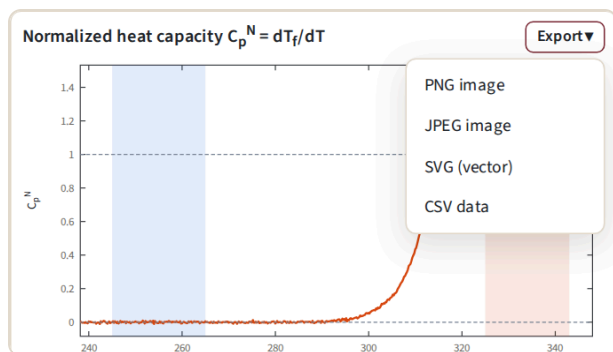


Figure 8.6. Export ▼ on a plot.

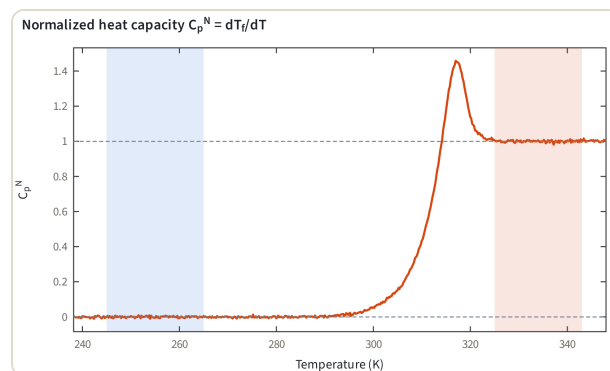


Figure 8.7. The exported PNG.

9 • Data files

- **Formats:** text files (.csv, .txt, .dat, .tsv) or pasted text. Columns may be separated by commas, semicolons, tabs or spaces.
- **Headers:** lines before the numbers (titles, column names, unit rows) are skipped; the last text line before the data is used for column names.
- **Decimal point:** use a period (0.25). Files with decimal commas must be converted first.
- **Order:** any; cooling scans may run from high to low temperature.
- **Instrument exports:** export to text/CSV from the instrument software (e.g. TA Universal Analysis / TRIOS, Netzsch Proteus, Mettler STARE) with temperature and heat flow columns.

Example (DSC, TA style)

Time (min),Temperature (°C),Heat Flow (W/g)

0.00,-120.0,-0.1000

0.05,-119.5,-0.1001

...

Example (annealing)

t (s) V (cm³/g)

1 0.795716

1.2 0.795715

...

10 • Troubleshooting and questions

Problem	What to do
The Fitter shows no model curves.	Expected: the model runs only when you click Compute or Run fit in step 4.
A plot shows “Could not draw ...”.	The rest of the page still works. Check the inputs of that part (e.g. a range with fewer than two points) and report it if it persists.
A hold shows no relaxation.	The hold is above T_g : the liquid equilibrates within milliseconds. Put the hold below T_g .
Mac: “cannot be opened / cannot verify the developer”.	System Settings → Privacy & Security → Open Anyway (once per computer).
Mac: “Restart now” does nothing, or the version does not change.	Download the latest disk image (.dmg) and replace the app (section 2.2). Versions before 1.1.6 cannot update themselves.
Windows: “Windows protected your PC”.	More info → Run anyway (first install only).
TL map or β_{KWW} plot missing.	Very long histories (above ~60 000 steps with 200 domains) do not keep every $T_{f,i}$. Use a coarser ΔT step or fewer hold points.
A fitted parameter is “at bound”.	Widen its bounds in step 3, or fix it if the data cannot determine it.
Parameters strongly correlated ($ r > 0.95$).	Add data that separate them (several cooling rates, annealing data) or fix one of them.
C_p^N is upside down.	Change the sign convention of the run (endothermic up/down).
A project is missing.	Projects are kept per browser and per app (section 8.5). Open the browser or app where you made it, or import its exported project file.
Save to project asks for a project name.	No project was open. Enter a name to create one and save into it, or cancel and open an existing project from Project ▼ .
ΔH has the wrong sign or size.	Check the sign convention, the rate and the sample mass, and that all scans used the same heating rate.

Report problems or ideas at github.com/wtakeda-research/GARASU/issues or to the Lucas group.

11 • Reference

Abbreviations

Abbreviation	Meaning
AIC, BIC	Akaike and Bayesian information criteria (fit quality penalized by the number of parameters; lower is better)
DSC	differential scanning calorimetry
HF, HF _g , HF _l	heat flow; extrapolated glass and liquid heat-flow lines
KWW	Kohlrausch–Williams–Watts (stretched exponential, $\exp[-(t/\tau)^\beta]$)
LM	Levenberg–Marquardt (local least-squares fitting)
MAP	Mauro–Allan–Potuzak nonequilibrium viscosity model (implemented in RelaxPy)
MDSC	modulated (temperature-modulated) differential scanning calorimetry, also TMDSC
MYEGA	Mauro–Yue–Ellison–Gupta–Allan viscosity equation
OTP, PVAc	<i>o</i> -terphenyl, poly(vinyl acetate) (material presets)
RMSE, SE	root-mean-square error, standard error
TL	Takeda–Lucas heterogeneous model
TNM	Tool–Narayanaswamy–Moynihan model
VFT	Vogel–Fulcher–Tammann equation

Model parameters

Model	Parameter	Meaning
TL	T_g	temperature where $\langle \tau \rangle = 100$ s
	m	fragility index
	$\log_{10} \tau_0$	high-temperature limit of τ (s)
	f	width of the relaxation-time distribution ($\approx \beta_{\text{KWW}}$ at T_g), 0.10–0.99
	β_0	intrinsic (local) stretching; 1 = fully heterogeneous
	N	number of domains (numerical; 200 is accurate)
TNM	T_g	$\tau(T_g) = 100$ s
	m	fragility; $\Delta h/R = m \ln 10 T_g$
	x	nonlinearity (0–1)
	β	KWW stretching exponent
MAP / RelaxPy	T_g	$\eta(T_g) = 10^{12}$ Pa·s
	m	fragility in the Mauro–Yue–Ellison–Gupta–Allan (MYEGA) viscosity equation
	$\log_{10} \eta_\infty$	high-temperature viscosity limit
	A, B, C	nonequilibrium viscosity branch; A from continuity at T_g by default
	p	ergodicity exponent (default 0.3082· m)
	$\log_{10} K_s$	modulus in $\tau = \eta/K_s$
	β	stretching of the Prony series

Fitter data types

Data type	Linked to	Notes
DSC — normalized C_p (dT_f/dT) vs T	ramp	already normalized (e.g. from Data Analysis)
DSC — C_p or heat flow vs T	ramp	raw; glass/liquid baselines fitted
Enthalpy vs T	ramp, MDSC	$H = a + bT + \Delta C_p T_f$ fitted
Volume / density / length vs T	ramp, MDSC	dilatometry
Fictive temperature T_f vs T	ramp, MDSC	
Annealing — normalized $\phi(t)$	hold	$\phi = (T_f - T)/(T_{f,0} - T)$
Annealing — $T_f(t)$	hold	
Annealing — enthalpy / volume / refractive index vs t	hold	$a + b \cdot T_f$ fitted
MDSC — C_p' (reversing), C_p''	MDSC	normalized; optional scale

Glossary

T_f	fictive temperature: the temperature at which the liquid would have the current structure
T_f'	limiting fictive temperature of the glass (frozen value after cooling)
C_p^N	normalized heat capacity, 0 in the glass and 1 in the liquid; equals dT_f/dT
$\phi(t)$	normalized relaxation function, $1 \rightarrow 0$ during annealing
KWW	stretched exponential $\exp[-(t/\tau)^\beta]$
δT_f	spread (standard deviation) of the local fictive temperatures in the TL model

12 • How to cite, and references

GARASU is free software under the MIT License: you may use, copy, modify and redistribute it, including in your own code, provided the copyright notice and license text are kept. © 2026 Arizona Board of Regents on behalf of the University of Arizona. Source and license: github.com/wtakeda-research/GARASU.

If GARASU contributes to your work, please cite:

- W. Takeda and P. Lucas, “GARASU: Glass Aging, Relaxation And Simulation Utility,” software, version 1.5.1, Zenodo (2026). doi:10.5281/zenodo.23091595
- W. Takeda and P. Lucas, “A model of heterogeneous undercooled liquid and glass accounting for temperature-dependent nonexponentiality and enthalpy fluctuation,” *J. Chem. Phys.* **160**, 174504 (2024). doi:10.1063/5.0196812
- W. Takeda and P. Lucas, “Relationship between enthalpy fluctuation and nonexponential relaxation in glass-forming liquids,” *Mater. Adv.* **7**, 5729 (2026). doi:10.1039/d6ma00442c

and, when you use them, the TNM and MAP / RelaxPy papers:

- A. Q. Tool, *J. Am. Ceram. Soc.* **29**, 240 (1946); O. S. Narayanaswamy, *J. Am. Ceram. Soc.* **54**, 491 (1971); C. T. Moynihan et al., *J. Am. Ceram. Soc.* **59**, 12 (1976); I. M. Hodge, *Macromolecules* **20**, 2897 (1987); G. W. Scherer, *J. Am. Ceram. Soc.* **69**, 374 (1986).
- J. C. Mauro, D. C. Allan and M. Potuzak, *Phys. Rev. B* **80**, 094204 (2009); X. Guo et al., *J. Am. Ceram. Soc.* **101**, 1169 (2018); J. C. Mauro and Y. Z. Mauro, *Physica A* **506**, 75 (2018); J. C. Mauro, R. J. Loucks and P. K. Gupta, *J. Am. Ceram. Soc.* **92**, 75 (2009); C. J. Wilkinson, Y. Z. Mauro and J. C. Mauro, *SoftwareX* **7**, 255 (2018).

Experiment references (Kovacs, Macedo–Napolitano, Hodge, Schawe, Hutchinson–Montserrat) are listed in Learn → Further reading.

GARASU • Lucas group, Department of Materials Science and Engineering, University of Arizona • github.com/wtakeda-research/GARASU